

# Situation Calculus

## Video Lesson

Duration: 00:09:05

## Slide Deck

**M10S02** (<https://canvas.tamu.edu/courses/430127/files/92570684?wrap=1>) 

## Lesson Transcript

In this lecture, we will look into situation calculus.

---

## Situation Calculus

Make propositional-logic-based planner scalable.

- Situations: logical *terms* indicating a state.  
Example: In situation  $S_0$  taking action  $a$  leads to situation  $S_1$ :

$$S_1 = \text{Result}(a, S_0).$$

- Fluents: *functions* and *predicates* that vary from one situation to the next.  
Example:  $\neg \text{Holding}(\text{Gold}_1, S_0), \text{Age}(\text{Wumpus})$

Other stuff: Atemporal/eternal predicates  $\text{Gold}(\text{Gold}_1)$ , empty actions  $\text{Result}([], s) = s$ , sequence of actions  $\text{Result}([a|\text{seq}], s) = \text{Result}(\text{seq}, \text{Result}(a, s))$ .

As we have seen in the previous lecture, propositional logic-based planning is very inefficient because we have to represent all of these different parametric values as part of the name of the atom.

Situation calculus was developed to address this particular issue. In situation calculus, there are two important concepts, situations and fluents.

Situations are logical terms indicating a particular state. For example, in situation  $S_0$  taking Action  $a$  leads to Situation  $S_1$ , which is expressed as the result of  $a$  in  $S_0$  is  $S_1$ . Fluents, on the other hand, are functions and predicates that vary from one situation to the next. For example, not holding  $\text{Gold}_1, S_0$  means that the agent is not holding  $\text{Gold}_1$  in situation  $S_0$ .  $\text{Age}(\text{Wumpus})$  means what the age of the Wumpus is. For example, whenever the agent takes one action, the age of the Wumpus would increase by one.

There is other stuff, such as atemporal or eternal predicates. For example, object  $\text{Gold}_1$  is a piece of gold.

You can also define the resulting situation starting from a situation followed by a sequence of actions. To do this, we first define an empty action. If you are in a situation,  $S$ , and if the action was empty, then the resulting situation is the original situation itself.

---

# Situation Calculus

Make propositional-logic-based planner scalable.

- Situations: logical *terms* indicating a state.

Example: In situation  $S_0$  taking action  $a$  leads to situation  $S_1$ :

$$S_1 = \text{Result}(a, S_0).$$

- Fluents: *functions* and *predicates* that vary from one situation to the next.

Example:  $\neg \text{Holding}(\text{Gold}_1, S_0)$ ,  $\text{Age}(\text{Wumpus})$

Other stuff: Atemporal/eternal predicates  $\text{Gold}(\text{Gold}_1)$ , empty actions  $\text{Result}([], s) = s$ , sequence of actions  $\text{Result}([a|\text{seq}], s) = \text{Result}(\text{seq}, \text{Result}(a, s))$ .



For example, if you want to know what is the resulting situation when you start from Situation  $S$  and follow Action  $a$  followed by a sequence of actions, then it can be recursively defined as the resulting situation where we execute a sequence of actions on the situation that results from applying action  $a$  to the initial Situation,  $S$ .

To look at this more clearly, let us consider this resulting Situation  $S'$ , which is illustrated below. You start from Situation  $S$  and apply Action  $a$ , and you end up in Situation  $S'$ . Following that, you execute a series of actions or a sequence of actions. In the end, you get to Situation  $S''$ .

When you write this out, this becomes the result followed by a sequence, starting from Situation  $S$ , which is exactly what this is. And it is what this is.

---

# Situation Calculus: Tasks

- Projection:  
Deduce the outcome of a given sequence of actions
- Planning:  
Find a sequence of actions that achieves a desired effect.  
Example: Wumpus world

Initial:  $At(Agent, [1, 1], S_0) \wedge At(G_1, [1, 2], S_0), \dots$   
Goal:  $\exists seq \ At(G_1, [1, 1], Result(seq, S_0))$

There are two basic tasks in situation calculus, projection and planning. Projection means deducing the outcome of a given sequence of actions. That is, how will all of these different fluents change as a consequence of taking these actions?

Planning, on the other hand, wants to find the sequence of actions that achieve the desired effect.

For example, in the Wumpus World, the initial situation is described as the agent being at location [1, 1] in Situation zero, the gold Object  $G_1$  being in location [1, 2] in Situation  $S_0$ , and so on. So, all of these would be fluents describing the state of the world in Situation  $S_0$ .

The goal can be described like this. There exists a sequence such that the goal Object  $G_1$  is at location [1, 1] in a situation resulting from following the sequence of actions "seq", starting from Situation  $S_0$ .

---

## Describing Actions in Situation Calculus

Two axioms:

- Possibility axiom: when it is possible to execute an action

$Preconditions \rightarrow Poss(a, s)$

- Effect axiom: What happens when a possible action is taken

$Poss(a, s) \rightarrow \text{Changes that result}$



There are two types of axioms that describe actions in situation calculus. The first is the possibility axiom. The second is the effect axiom.

Possibility axioms tell you when it is possible to execute an action. So, if these pre-conditions are met, then it is possible to take Action  $a$  in Situation  $s$ .

Effect axioms, on the other hand, tell you what happens when a possible action is taken. So, the left-hand side of the implication now becomes, it is possible to take action in Situation  $s$ . Then what are the changes that result? So, these would indicate which fluents change.

---

## Wumpus World: Axioms

- Possibility axioms: Move, grab, release

$$At(Agent, x, s) \wedge Adjacent(x, y) \rightarrow Poss(Go(x, y), s)$$

$$Gold(g) \wedge At(Agent, x, s) \wedge At(g, x, s) \rightarrow Poss(Grab(g), s)$$

$$Holding(g, s) \rightarrow Poss(Release(g), s)$$

- Effect axioms: Move, Grab, Release

$$Poss(Go(x, y), s) \rightarrow At(Agent, y, Result(Go(x, y), s))$$

$$Poss(Grab(g), s) \rightarrow Holding(g, Result(Grab(g), s))$$

$$Poss(Release(g), s) \rightarrow \neg Holding(g, Result(Release(g), s))$$

Let us look at some "possibility" axioms and "effect" axioms in the Wumpus World domain.

Let us look at these three possibility axioms: move, grab, and release. Each means, is it possible to move? Is it possible to grab? Is it possible to release? First is the move possibility axiom. So, if the agent is at location  $x$  in situation  $s$ , and these two locations,  $x$  and  $y$ , are adjacent to each other, then it is possible to go from  $x$  to  $y$  in that situation  $s$ .

Next is the possibility of grabbing something. Here it says if this Object  $g$  is a piece of gold and if the agent is at location  $x$  in situation  $s$ , and this Object  $g$  is also at location  $x$  in Situation  $s$ , then it is possible to grab  $g$  in that Situation  $s$ .

If you did not have this fluent to tell you that this Object  $g$  is a piece of gold, and if you only had that, then perhaps it would be possible to grab a wumpus, and that would be fatal. And why does this gold fluent not have the situation? This is because gold is an eternal fluent, meaning that regardless of the particular situation you are in, the property of this object, if it is gold, it does not change.

Next, let us look at the possibility of releasing something. So, if in a Situation  $s$  you are holding onto some Object  $g$ , then it is possible to release this Object  $g$  in the same situation.

Now let us consider effect axioms. Here we will consider move, grab, and release. That is, what are the effects of moving, grabbing, and releasing? First, it must be possible to go from  $x$  to  $y$  in a Situation  $s$ . Then you can think about what happens when you actually go from  $x$  to  $y$ . So, when it is possible, then in the situation that results from executing this action, go from  $x$  to  $y$ , then the agent will now be in location  $y$ .

The second example. If it is possible to grab Object  $g$  in Situation  $s$ , then in the resulting situation, we actually grab  $g$  in the original Situation  $s$ . This could be gold or any object. So, as a result of grabbing, you will be holding onto that in this resulting situation.

Finally, what would be the effect of releasing? First, it must be possible to release that Object  $g$  in Situation  $s$ , and when that is satisfied. And when you execute that action in the original Situation  $s$ , then in the resulting situation, you will not be holding that Object  $g$ .

In the following lecture, we will look at the issues that arise from this formalism and how to solve those issues.